### 11.2.10 Hormones of heart, kidney and gastro intestinal tract

Some tissues of the heart, kidney and gastro intestinal tract acts as partial endocrine glands. In the heart, cardiocytes on the atrial wall's secretes an important peptide hormone called atrial natriuretic factor (ANF). When blood pressure is increased, ANF is secreted and causes dilation of the blood vessels to reduce the blood pressure.

In kidneys, hormones such as renin, erythropoietin and calcitriol are secreted. **Renin** is secreted by juxta glomerular cells (JGA), which increases blood pressure when angiotensin is formed in blood. **Erythropoietin** is also secreted by the JGA cells of the kidney and stimulates erythropoiesis (formation of RBC) in bone marrow. **Calcitriol** is sercreted by proximal tubules of nephron. It is an active form of vitamin $D_3$ which promotes calcium and phosphorus absorption from intestine and accelerates bone formation.

#### Gastro intestinal tract hormones

Group of specialized endocrine cells present in gastro-intestinal tract secretes hormones such as gastrin, cholecystokinin (CCK), secretin and gastric inhibitory peptides (GIP). **Gastrin** acts on the gastric glands and stimulates the secretion of HCl and pepsinogen. **Cholecystokinin (CCK)** is secreted by duodenum in response to the presence of fat and acid in the diet. It acts on the gall bladder to release bile into duodenum and stimulates the secretion of pancreatic enzymes and its discharge. **Secretin** acts on acini cells of pancreas to secrete bicarbonate ions and water to neutralize the acidity. **Gastric inhibitory peptide (GIP)** inhibits gastric secretion and motility.

## 11.3 Hypo and Hyper activity of endocrine glands and related disorders

The hyper secretion and hypo secretion of hormones leads to several disorders

**Dwarfism** is due to hyposecretion of growth hormone (GH) in children, skeletal growth and sexual maturity is arrested. They attain a maximum height of 4 feet only (Figure 11.8).



**Figure 11.8** Dwarfism

**Gigantism** is due to hypersecretion of growth hormone (GH) in children. Overgrowth of skeletal structure occurs (up to 8 feet) and the visceral growth is not appropriate with that of limbs. Figure 11.9.

**Acromegaly** is due to excessive secretion of growth hormone in adults. Over growth of hand bones, feet bones, jaw bones, malfunctioning of gonads, enlargement of viscera, tongue, lungs, heart, liver, spleen and endocrine gland like thyroid, adrenal etc., are the symptoms of acromegaly. (Figure 11.10)



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